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| **Date** | 07-07-2026 |
| **Team ID** | |
| **Project Name** | Automated Credit Card Approval Prediction using Machine Learning |
| **Maximum Marks** | 2 Marks |

# Technology Stack

### Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs used to build the solution. A well-chosen technology stack ensures that the application is scalable, maintainable, and aligned with the project requirements.

Below is a detail of the specific technologies chosen for each component of the architecture, with a brief justification for each choice.

| S.No | Architecture Component / Layer | Technology Chosen | Justification / Purpose |
|---|---|---|---|
| 1 | **Frontend / ClientSide** | HTML5, CSS3 (customstyled, no framework) | HTML and CSS provide a simple and responsive user interface, while Bootstrap improves the design and responsiveness. JavaScript is used for client-side validation and interactive user experience. |
| 2 | **Backend / ServerSide** | Python + Flask | Python is used because the Machine Learning model is developed using Scikit-learn. Flask provides a lightweight web framework for handling requests, processing user input, and displaying prediction results. |
| 3 | **Database / Data Storage** | Static CSV file (Crop_recommendation.csv) + Pickle (.pkl) files | Applicant data for model training is stored in a CSV dataset. The trained Machine Learning model and preprocessing objects are saved as Pickle files and loaded during prediction for fast execution. |

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| 4 | **Cloud / Hosting / Deployment** | Render (free tier) | Render provides an easy deployment platform for Flask applications directly from GitHub without requiring additional server configuration. |
| 5 | **Version Control & CI/CD** | Git + GitHub | Git is used for version control, while GitHub stores the project repository and enables collaboration, backup, and deployment integration. |
| 6 | **Third-Party APIs / Other Tools** | None currently integrated | Scikit-learn is used for model development, Pandas and NumPy for data preprocessing and analysis, and Joblib for saving/loading trained models efficiently. |